A/B TESTING AND HYPOTHESIS TESTING REPORT

Checkout Conversion Experiment

Dataset: checkout_ab_test_v2.csv
Report generated: 11 October 2026

1. Objective

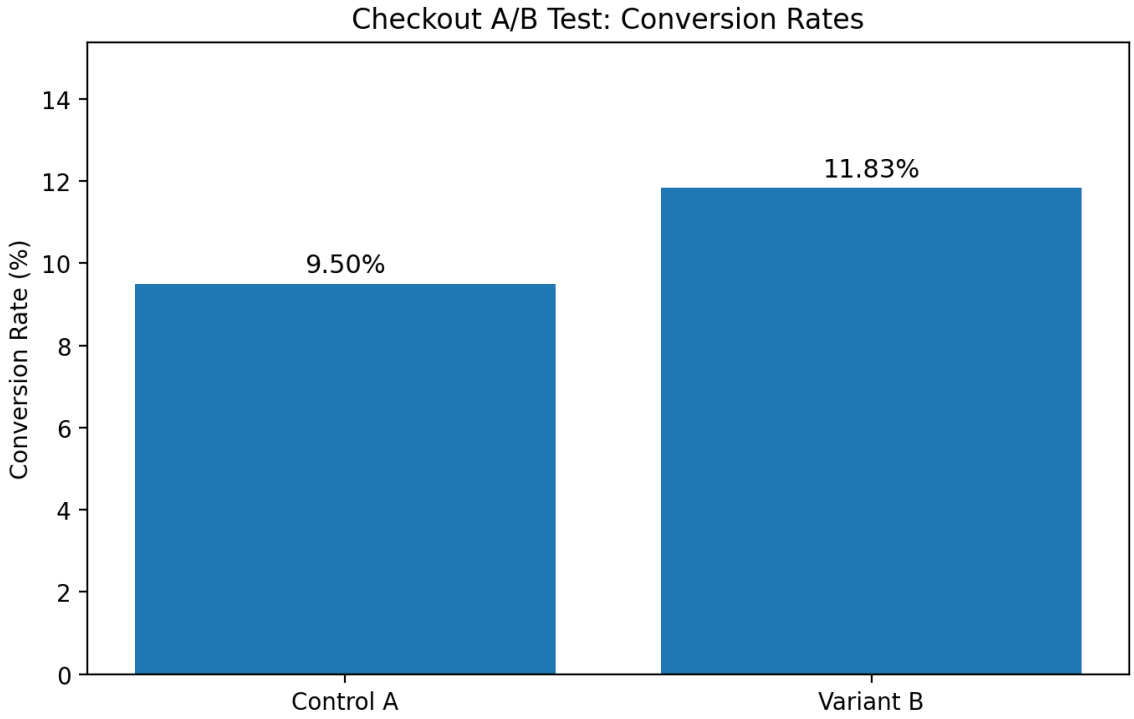
Evaluate whether the modified checkout experience (Variant B) produces a different conversion rate from the existing checkout experience (Control A).

2. Data Quality

The dataset contains 1200 rows and 8 columns, with 1200 unique visitors. Total missing values across all columns: 0.

3. Conversion Results

Metric	Control A	Variant B
Visitors	600	600
Conversions	57	71
Non-conversions	543	529
Conversion rate	9.50%	11.83%



4. Hypotheses and Statistical Test

Null hypothesis (H0): The population conversion rates are equal.
Alternative hypothesis (H1): The population conversion rates differ.
A two-sided two-proportion Z-test was conducted at a significance level of 0.05.

Statistical measure	Result
Absolute lift (B - A)	2.33%
Relative lift	24.56%
Z-statistic	1.3092
P-value	0.1905
95% confidence interval	[-1.16%, 5.82%]
Cohen's h	0.0757
Significance level	0.05

5. Interpretation

Variant B converted 71 of 600 visitors (11.83%), compared with 57 of 600 visitors (9.50%) for Control A. The observed absolute lift was 2.33%, and the relative lift was 24.56%. The p-value was 0.1905, which is above the 0.05 significance level. The 95% confidence interval [-1.16%, 5.82%] includes zero. Therefore, the result is: No statistically significant difference.

6. Executive Decision Memo

Objective: Determine whether the modified checkout experience should replace the existing checkout page.

Finding: Variant B recorded a higher observed conversion rate than Control A, but the difference was not statistically significant at the 5% level.

Recommendation: Do not declare Variant B the winner yet. Continue testing with an adequately planned sample size.

Further work: Estimate an appropriate sample size for a follow-up experiment and investigate results by device type and traffic source before making a final rollout decision.

7. Conclusion

Variant B showed a higher observed conversion rate, but the experiment does not provide sufficient evidence of a difference at the 5% significance level. The null hypothesis cannot be rejected. Further testing is recommended before declaring Variant B the winner.